

Getting Started

Thank you for choosing Freenove products!

This tutorial has two parts.

The part 1 is for Freenove Raspberry Pi starter or case kit. It will introduce how to assemble case and setup raspberry Pi.

The part 2 is for Raspberry Pi projects. If you have electronic components, you can learn our tutorial.

Get Support and Offer Input

Freenove provides free and responsive product and technical support, including but not limited to:

- Product quality issues
- Product use and build issues
- Questions regarding the technology employed in our products for learning and education
- Your input and opinions are always welcome
- We also encourage your ideas and suggestions for new products and product improvements

For any of the above, you may send us an email to:

support@freenove.com

Safety and Precautions

Please follow the following safety precautions when using or storing this product:

- Keep this product out of the reach of children under 6 years old.
- This product should be used only when there is adult supervision present as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and parts, which are sharp. This product contains electrically conductive parts. Use caution with electrically conductive parts near or around power supplies, batteries and powered (live) circuits.
- When the product is turned ON, activated or tested, some parts will move or rotate. To avoid injuries to hands and fingers, keep them away from any moving parts!
- It is possible that an improperly connected or shorted circuit may cause overheating. Should this happen, immediately disconnect the power supply or remove the batteries and do not touch anything until it cools down! When everything is safe and cool, review the product tutorial to identify the cause.
- Only operate the product in accordance with the instructions and guidelines of this tutorial, otherwise parts may be damaged or you could be injured.
- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi and micro: bit
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

You can find more about Freenove and get our latest news and updates through our website:

<http://www.freenove.com>

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Part 1 Assemble Case and Set Up Raspberry Pi

This part is for **Freenove Starter Kit for Raspberry Pi** and **Freenove Case Kit for Raspberry Pi**. It will introduce how to assemble the case and setup your Raspberry Pi to get it work.

You can get the products via below:

<http://www.freenove.com/store.html>

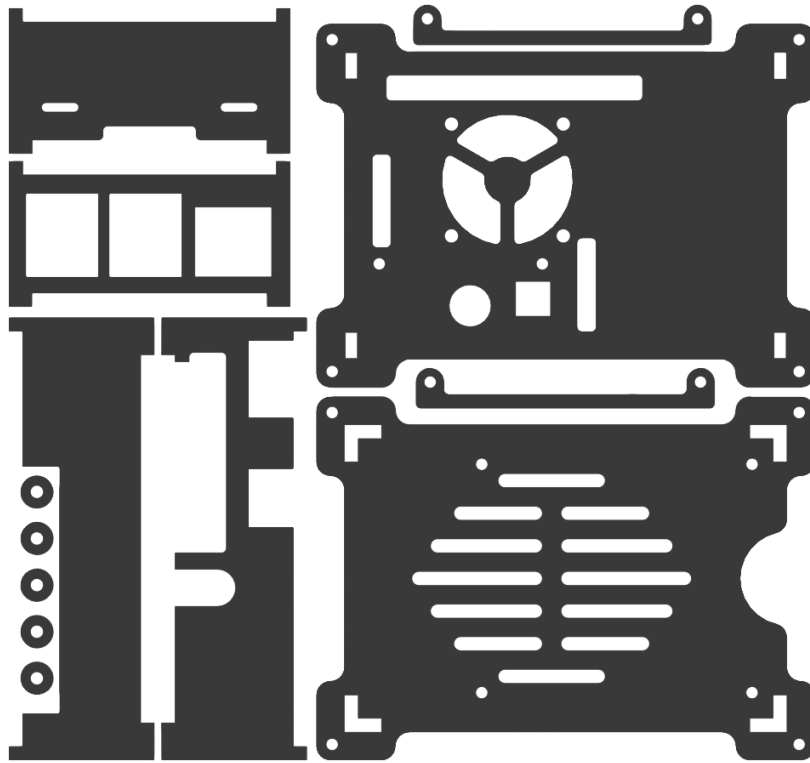
If you have any concerns, please feel free to contact us:

support@freenove.com

List

Acrylic and electronic parts

Acrylic



32 GB Samsung EVO (not contained in case kit)



Card Reader (not contained in case kit)



5V/3A Power Adapter (you get only one of below, which is applied in your country) (not contained in case kit)



MINI-HDMI Cable (not contained in case kit) 	USB-Type-C cable 10cm 
Power Fan board 	USB-Type-C cable 100cm (not contained in case kit V1, contained in V2) 
Fan & Screws and nuts for Fan 	Heat sink 

Mechanical Parts

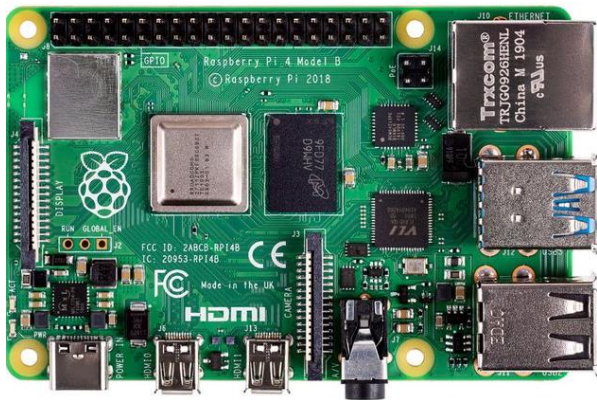
M2.5*6 Screw  x14 Freenove	M2.5*10 Screw  x5 Freenove	M2.5 Nut  x5 Freenove
M2.5*25 Copper Standoff  x5 Freenove	M2.5*8 Copper Standoff  x3 Freenove	Screwdriver 

Assemble Case

Note this case supports Raspberry Pi 4 B. It can be used for 3B and 3B + when without the side acrylic board for power port.

Step 1, paste heat sinks

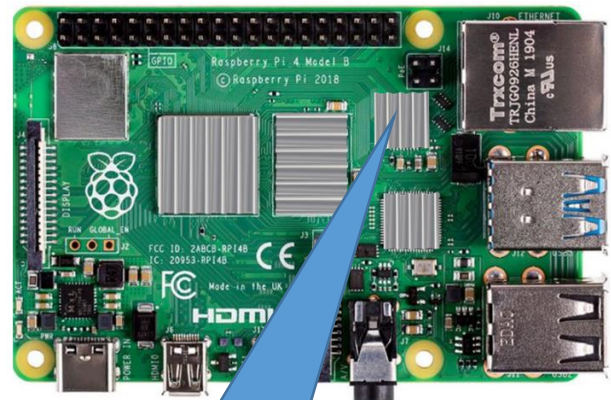
Before



Tear the cover of sink and paste on Raspberry Pi



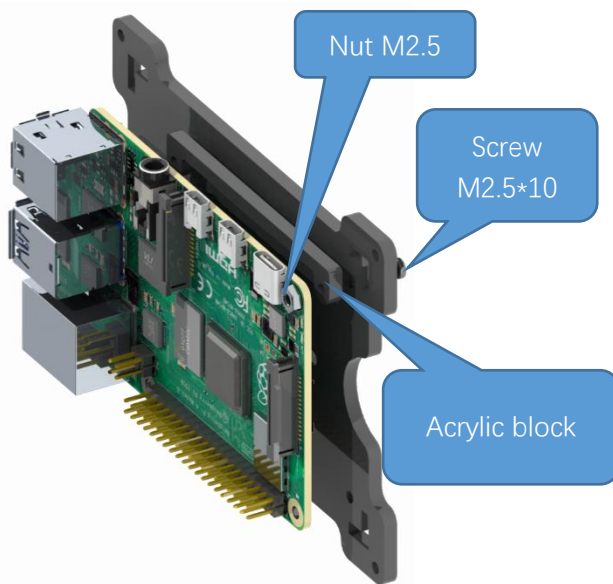
After



The two small heat sinks are not necessary.

Step 2, assemble the case

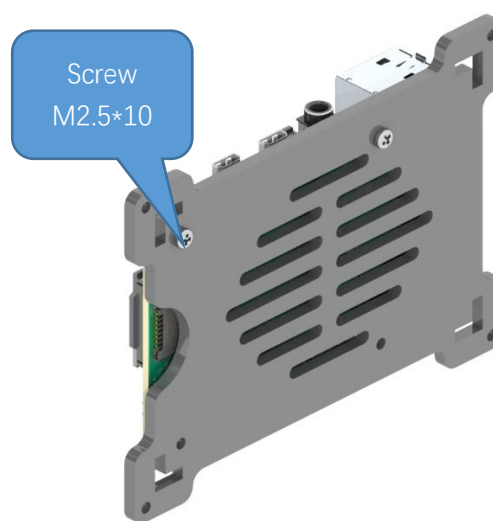
Install one side of RPi.



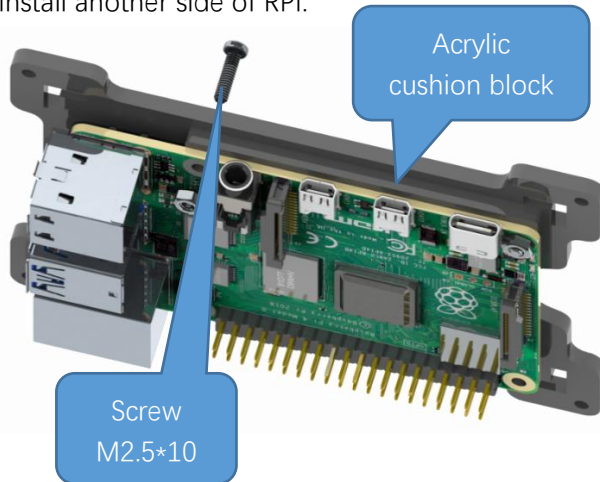
These parts are spare:



Bottom view.



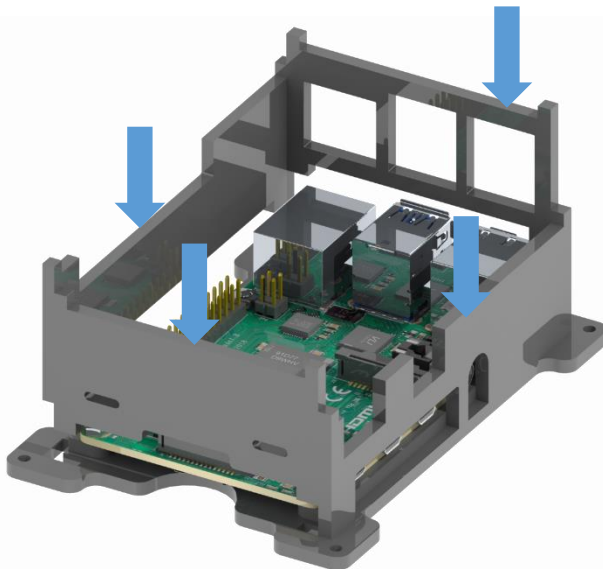
Install another side of RPi.



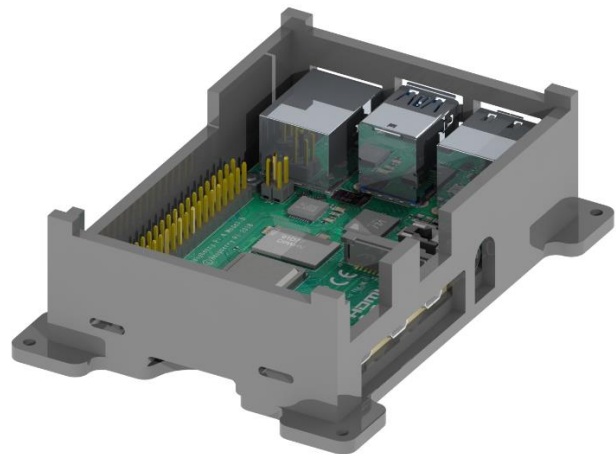
Raspberry pi is install to bottom board successfully.



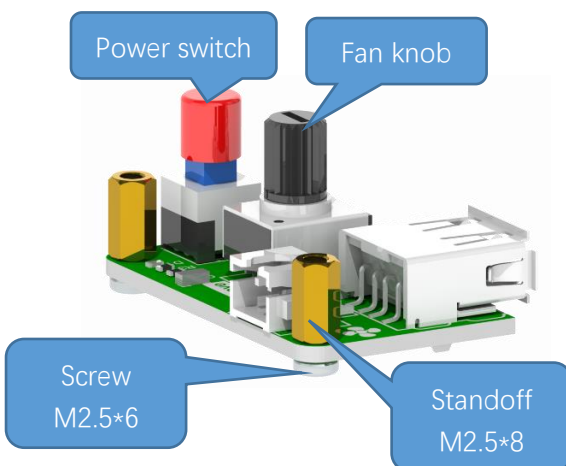
Install sideboard.



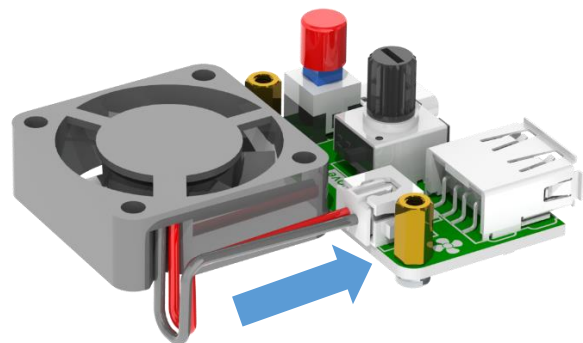
After installation.



Install standoff to power board.



Connect fan cable to power board.

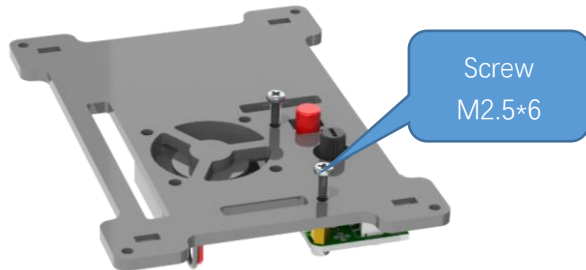


Install power board to top acrylic board.

1, Don't reverse fan.

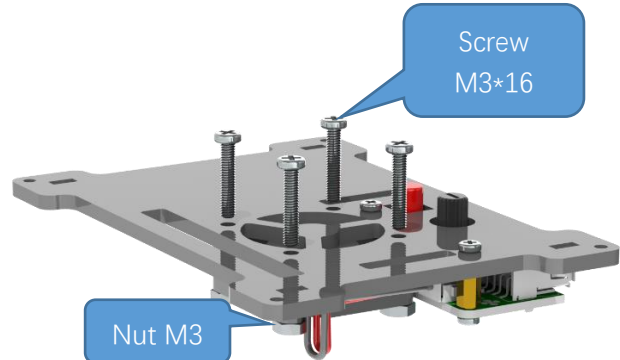
2, If you are using other heat sink, make sure your heat sink is shorter than ours.

Or it will influence fan.

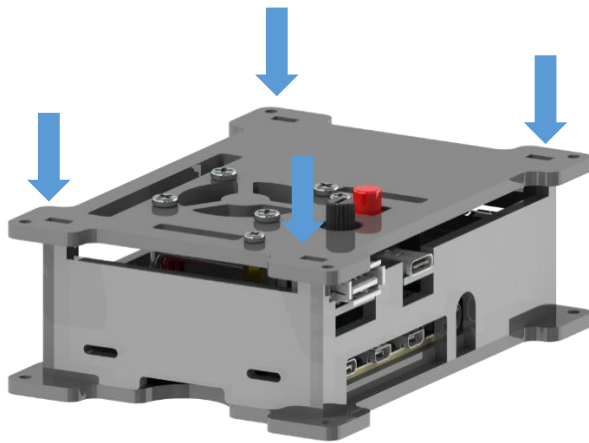


Install fan to top Acrylic board.

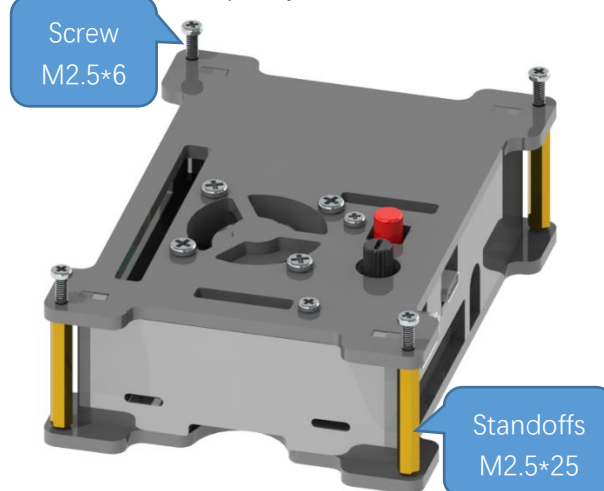
Plug 4 screws first. Then install Nuts.



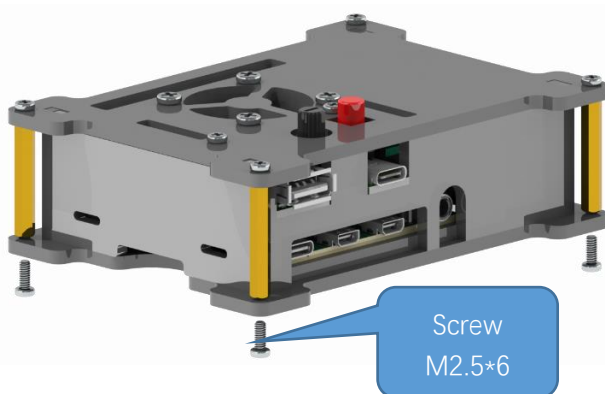
Install top acrylic board.



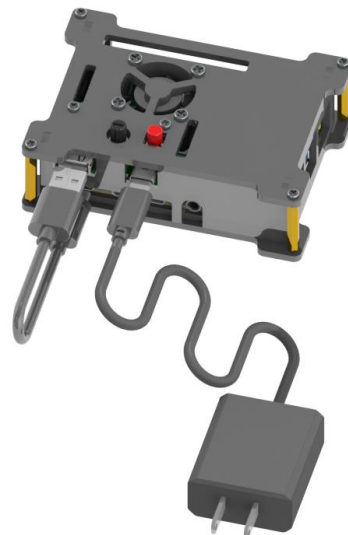
Install standoff to top acrylic board.



Install standoff to bottom acrylic board.



Next, you need to connect cables as below.



If you have any concerns (like broken parts or loud fan), please contact us:

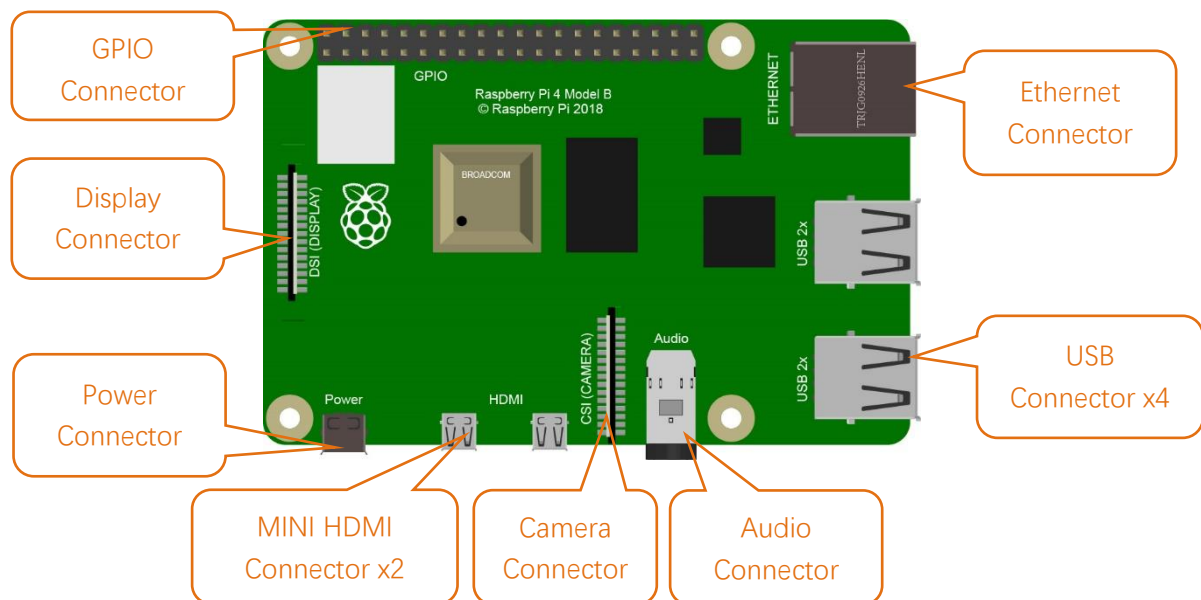
support@freenove.com

Raspberry Pi

Raspberry Pi (RPi, RPI, RasPi, these words will be used later), is a micro-computer with size of a card. When you learn it. You can take it as a computer. It is widely used in desktop workstation, media center, smart home, robots, and even the servers, etc.

So far, Raspberry Pi has developed to the fourth generation. Changes in versions are accompanied by increase and upgrades in hardware. A type and B type, the first generation of products, have been stopped due to various reasons. Other versions are popular and active and the most important is that they are consistent in the order and number of pins, which makes the compatibility of peripheral devices greatly enhanced between different versions.

Hardware interface diagram of RPi 4B is shown below:

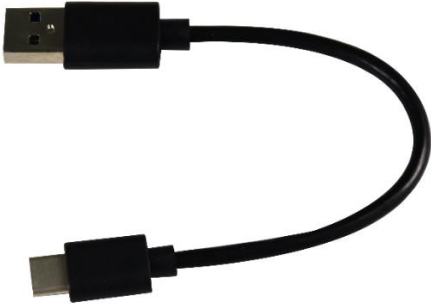
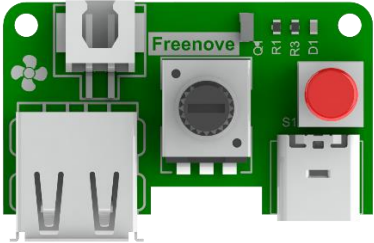


Install the System

Firstly, install a system for your RPi.

Component List

Necessary Components for this case

Any Raspberry Pi 	5V/3A Power Adapter. Choose the one applied in your country. 
USB-Type-C 10cm 	Micro SD Card(TF Card), Card Reader 
USB-Type-C cable 100cm  If you get Freenove USB-Type-C cable 100cm, you need use it. Or something may be wrong.	Power Fan board 

Power requirement of different versions of Raspberry Pi is shown in following table:

Product	Recommended PSU current capacity	Maximum total USB peripheral current draw	Typical bare-board active current consumption
Raspberry Pi Model A	700mA	500mA	200mA
Raspberry Pi Model B	1.2A	500mA	500mA
Raspberry Pi Model A+	700mA	500mA	180mA
Raspberry Pi Model B+	1.8A	600mA/1.2A (switchable)	330mA
Raspberry Pi 2 Model B	1.8A	600mA/1.2A (switchable)	350mA
Raspberry Pi 3 Model B	2.5A	1.2A	400mA
Raspberry Pi 3 Model A+	2.5A	Limited by PSU, board, and connector ratings only.	350mA
Raspberry Pi 3 Model B+	2.5A	1.2A	500mA
Raspberry Pi 4 Model B	3.0A	1.2A	600mA
Raspberry Pi Zero W	1.2A	Limited by PSU, board, and connector ratings only.	150mA
Raspberry Pi Zero	1.2A	Limited by PSU, board, and connector ratings only	100mA

For more details, please refer to <https://www.raspberrypi.org/help/faqs/#powerReqs>

In addition, RPi also needs a network cable used to connect it to wide area network.

All of these components are necessary. Among them, the power supply is required at least 5V/2.5A, because lack of power supply will lead to many abnormal problems, even damage to your RPi. So power supply with 5V/2.5A is highly recommend. SD Card Micro (recommended capacity 16GB or more) is a hard drive for RPi, which is used to store the system and personal files. In later projects, the components list with a RPi will contains these required components, using only RPi as a representative rather than presenting details.

Optional Components

Under normal circumstances, there are two ways to login to Raspberry Pi: using independent monitor, or remote desktop to share a monitor with your PC.

Required Accessories for Monitor

If you want to use independent monitor, mouse and keyboard, you also need the following accessories.

1. Display with HDMI interface
2. Mouse and Keyboard with USB interface

As to Pi Zero and Pi Zero W, you also need the following accessories.

1. HDMI to HDMI converter wire.
2. Micro-USB to USB-A Receptacles converter wire (Micro USB OTG wire).
3. USB HUB.
4. USB transferring to Ethernet interface or USB Wi-Fi receiver.

For different Raspberry Pi, the optional items are slightly different. But all of their aims are to convert the special interface to standard interface of standard Raspberry Pi.

	Pi Zero	Pi Zero W	Pi A+	Pi 3A+	Pi B+/2B	Pi 3B/3B+/4B
Monitor	Yes	Yes	Yes	Yes	Yes	Yes
Mouse	Yes	Yes	Yes	Yes	Yes	Yes
Keyboard	Yes	Yes	Yes	Yes	Yes	Yes
Mini-HDMI to HDMI converter wire	Yes	Yes	No	No	No	No
Micro-USB to USB-A Receptacles converter wire (Micro USB OTG wire)	Yes	Yes	No	No	No	No
USB HUB	Yes	Yes	Yes	Yes	No	No
USB transferring to Ethernet interface	select one from two	optional	select one from two	optional	Internal Integration	Internal Integration
USB Wi-Fi receiver	or select two from two	Internal Integration	or select two from two	Internal Integration	optional	

Required Accessories for Remote Desktop

If you do not have an independent monitor, or you want to use a remote desktop, first you need to login to Raspberry Pi through SSH, and then open the VNC or RDP service. So you need the following accessories.

	Pi Zero	Pi Zero W	Pi A+	Pi 3A+	Pi B+/2B	Pi 3B/3B+/4B
Micro-USB to USB-A Receptacles converter wire (Micro USB OTG wire)	Yes	Yes	No	NO		
USB transferring to Ethernet interface	Yes	Yes	Yes			

Raspbian System

If you are using MacBook, please refer to following page. Then jumper to next [Remote desktop & VNC](https://projects.raspberrypi.org/en/projects/raspberry-pi-setting-up).
<https://projects.raspberrypi.org/en/projects/raspberry-pi-setting-up>

If you are using Windows system, please follow content below:

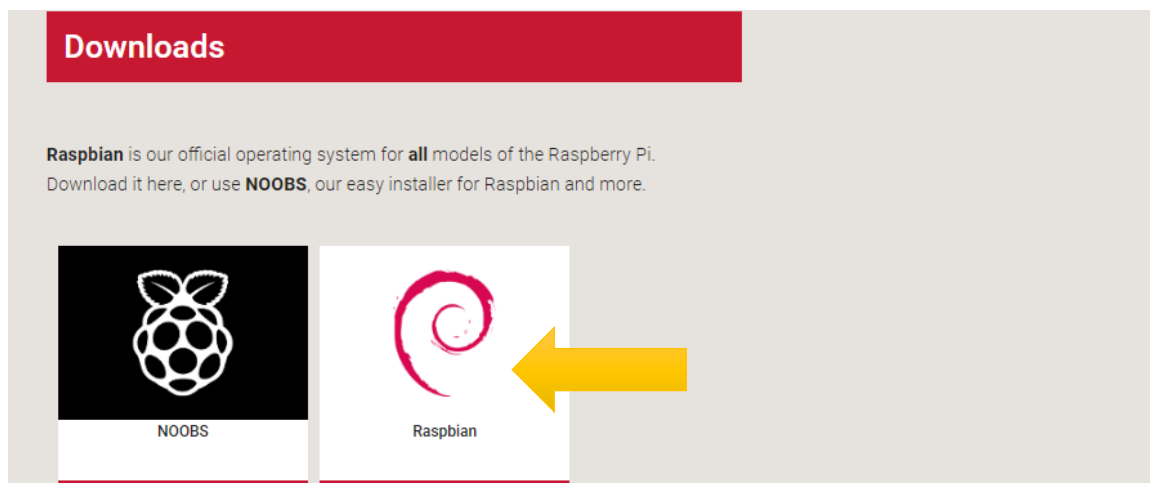
Tool and System image

Software Tool

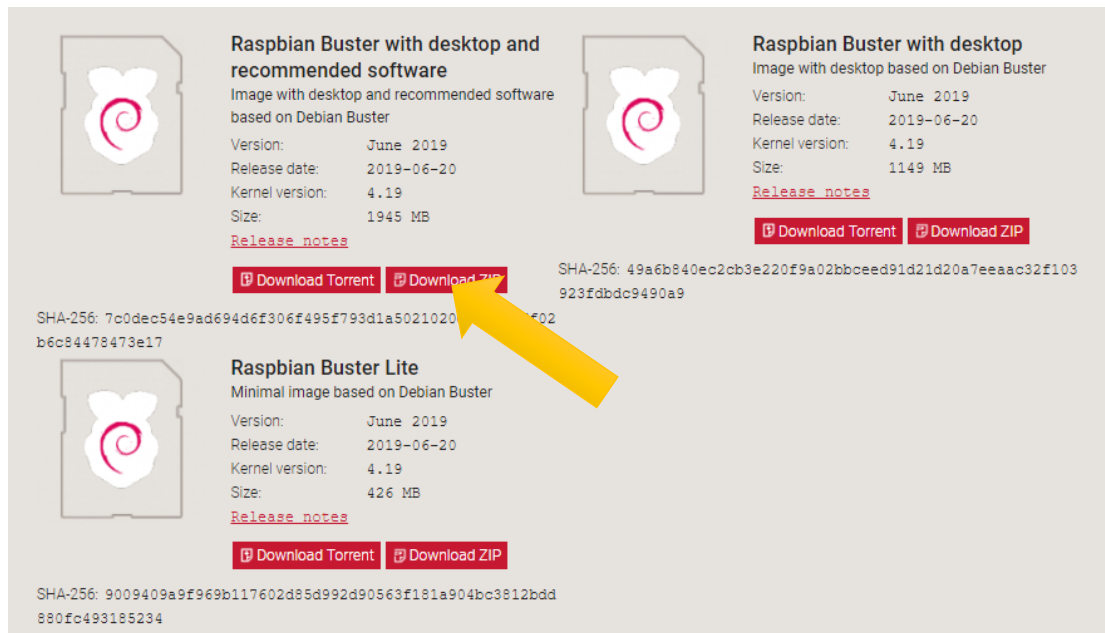
A tool Disk Imager Win32 is required to write system. You can download and install it through visiting the web site: <https://sourceforge.net/projects/win32diskimager/>

Selecting System

Visit RPi official website (<https://www.Raspberrypi.org/>), click “Downloads” and choose to download “RASPBIAN”. RASPBIAN supported by RPi is an operating system based on Linux, which contains a number of contents required for RPi. We recommended RASPBIAN system to beginners. All projects in this tutorial are operated under the RASPBIAN system.



<https://www.raspberrypi.org/downloads/raspbian/>



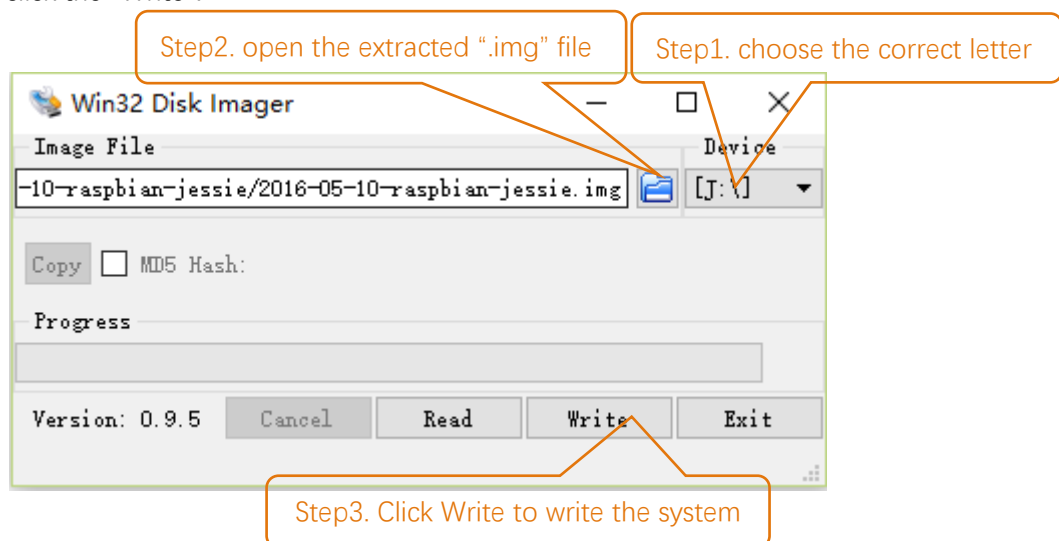
After download, extract file with suffix (.img). Preparation is ready to start making the system.

Write System to Micro SD Card

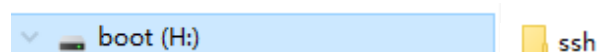
First, put your Micro SD card into card reader and connect card reader to USB port of your computer.



Then open Win32 disk imager, choose the correct letter of your Micro SD Card (here is "J"), open the extracted ".img" file and then click the "Write".



After the system is written. Create a new folder named "ssh" under generated boot disk.



If you have any concerns, please feel free to contact us: support@freenove.com

Start Raspberry Pi

If you have any concerns, please feel free to contact us: support@freenove.com

If you don't have a spare display, you can jumper to next section "[Remote desktop & VNC](#)".

If you have a spare monitor, please follow steps below.

After the system is written successfully, take out Micro SD Card and put it into the card slot of RPi. Then connect RPi to screen through the HDMI-micro HDMI, to mouse and keyboard through the USB port, to network cable through the network card interface and to the power supply. Then your RPi starts initially. Later, you need to enter the user name and password to login. The default user name: pi; password: raspberry. Enter and login. After login, you can enter the following interface.



Now, you have successfully installed the RASPBIAN operating system for your RPi.

Remote desktop & VNC

If you don't have a spare display, mouse and keyboard for your RPi, you can use a remote desktop to share a display, keyboard, and mouse with your PC. Below is how to use remote desktop to control RPi under the Windows operating system.

Under windows, Raspberry Pi can be generally accessed remotely through two applications. The first one is the windows built-in application remote desktop, which corresponds to the Raspberry Pi xrdp service. The second one is the free application VNC Viewer, which corresponds to the VNC interface of Raspberry Pi. Each way has its own advantages. You can choose either one or two.

Windows	Raspberry Pi
Remote Desktop Connection	Xrdp
VNC Viewer	VNC

VNC Viewer can not only run under Windows, but also under system MAC, Linux, IOS, Android and so on.

SSH

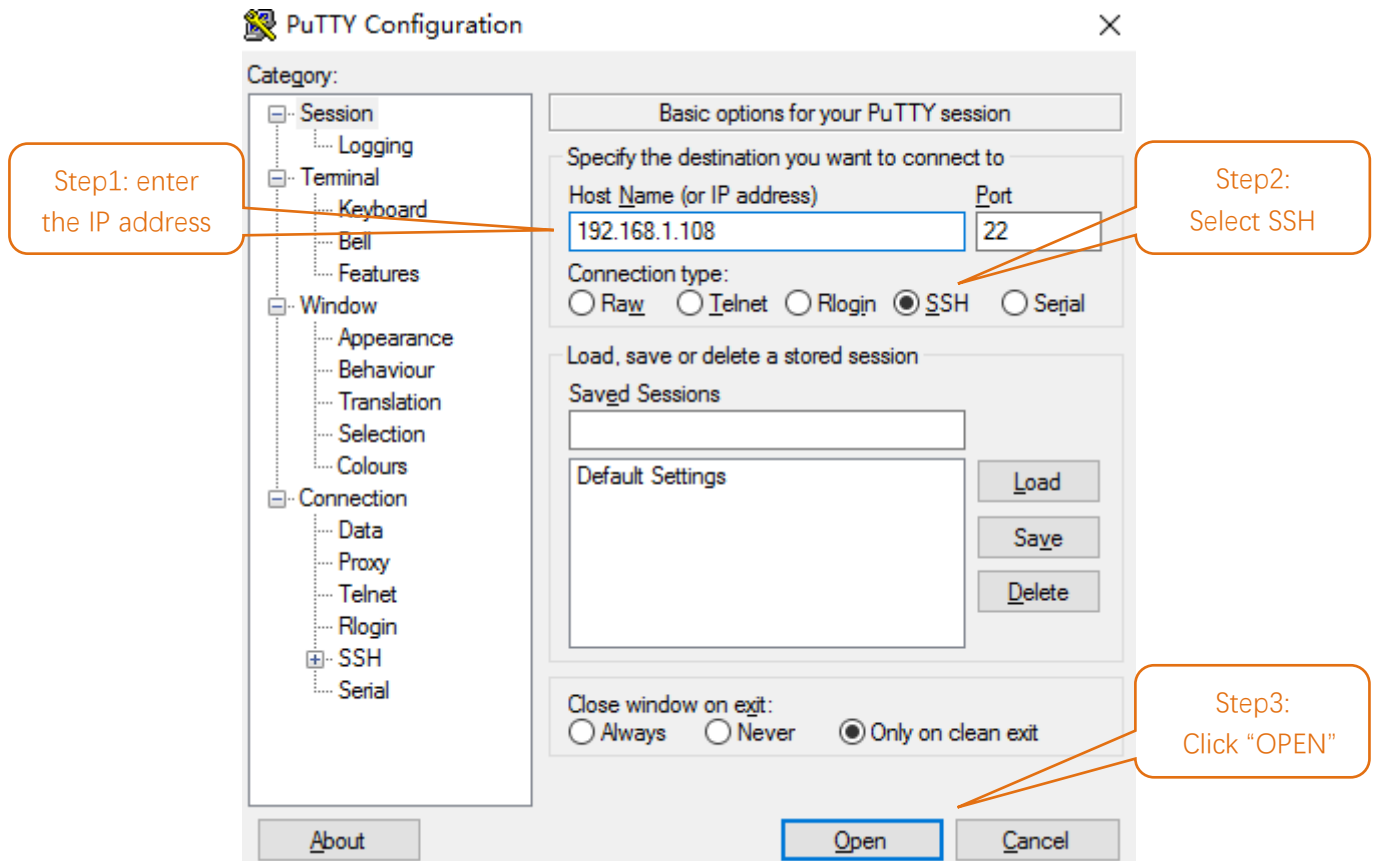
Under previous Raspbian system, SSH is opened by default. Under the latest version of Raspbian system, it is closed by default. So you need to open it first.

Method: after the system is written. Create a folder named “ssh” under generated boot disk, then the SSH connection will be opened.

Then, download the tool software Putty. Its official address: <http://www.putty.org/>

Or download it here: <http://www.chiark.greenend.org.uk/~sgtatham/putty/download.html>

Then use cable to connect your RPi to the routers of your PC LAN, to ensure your PC and your RPi in the same LAN. Then put the system Micro SD Card prepared before into the slot of the RPi and turn on the power supply waiting for starting RPi. Later, enter control terminal of the router to inquiry IP address named "raspberrypi". For example, I have inquired to my RPi IP address, and it is "192.168.1.108". Then open Putty, enter the address, select SSH, and then click "OPEN", as shown below:



There will appear a security warning at first login. Just click "YES".

PuTTY Security Alert



WARNING - POTENTIAL SECURITY BREACH!

The server's host key does not match the one PuTTY has cached in the registry. This means that either the server administrator has changed the host key, or you have actually connected to another computer pretending to be the server.

The new rsa2 key fingerprint is:

ssh-rsa 2048 7a:e1:50:ba:dc:01:87:1b:a5:f9:d2:d4:12:d6:fe:ab

If you were expecting this change and trust the new key, hit Yes to update PuTTY's cache and continue connecting.

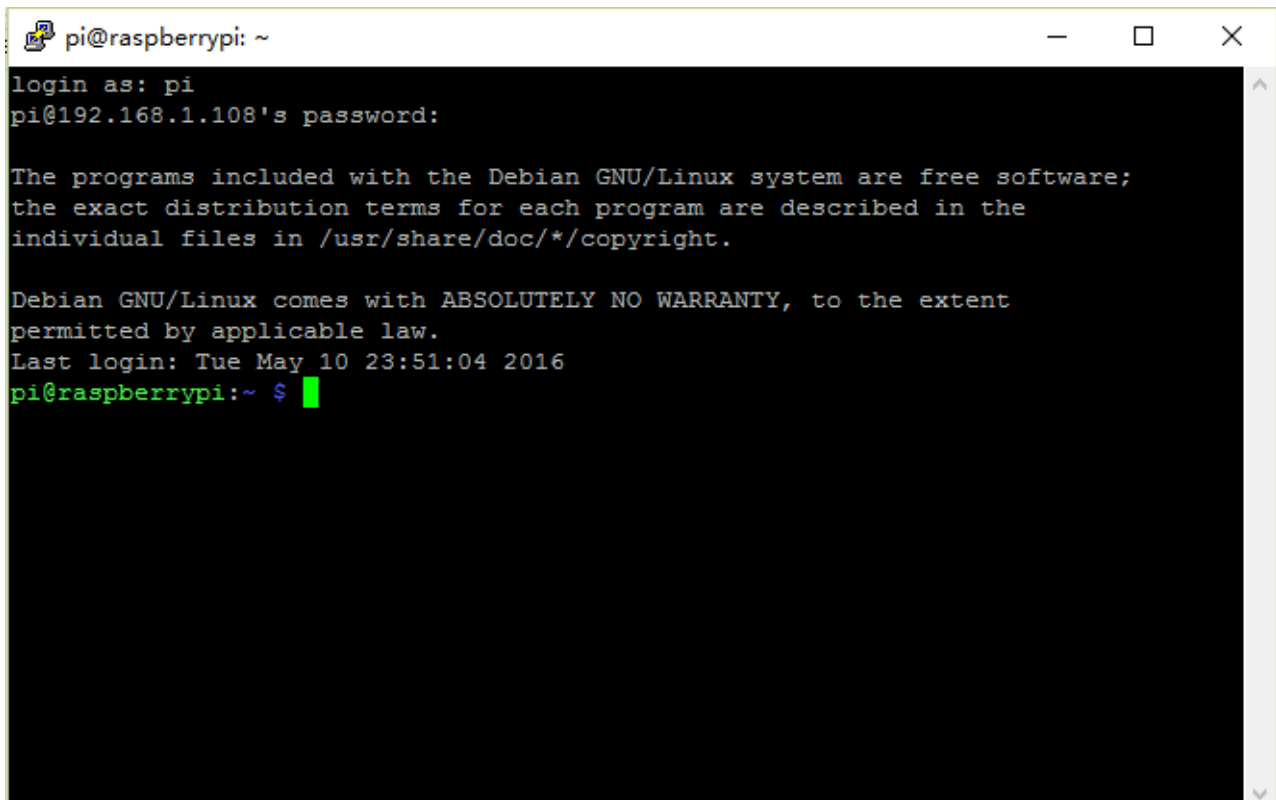
If you want to carry on connecting but without updating the cache, hit No.

If you want to abandon the connection completely, hit Cancel. Hitting Cancel is the ONLY guaranteed safe choice.

Then there will be a login interface (RPi default user name: pi; the password: raspberry). When you enter the password, there will be no display on the screen. This is normal. After the correct output, press “Enter” to confirm.



Then enter the command line of RPi, which means that you have successfully login to RPi command line mode.



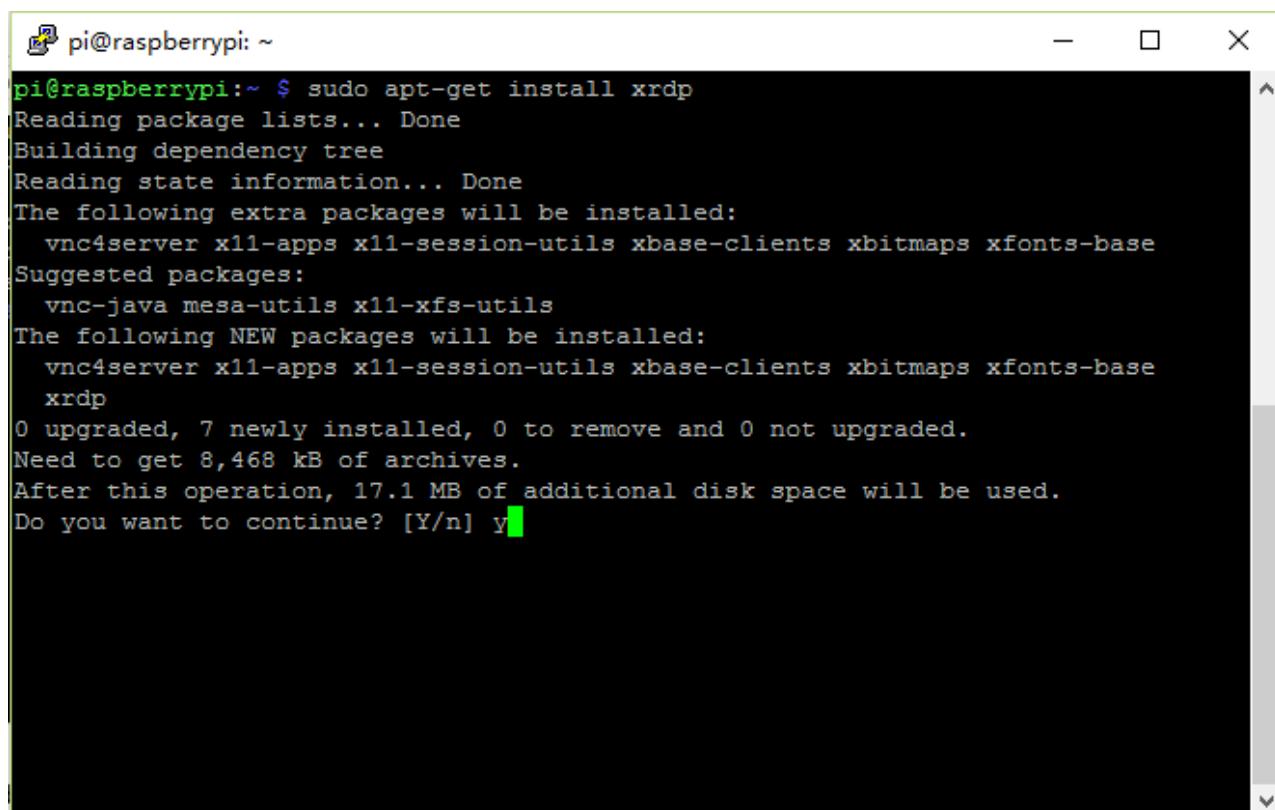
Remote Desktop Connection & xrdp

If you want to use built-in Remote Desktop Connection under Windows, you need install xrdp service on Raspberry Pi.

Next, install a xrdp service, an open source remote desktop protocol(rdp) server, for RPi. Type the following command, then press enter to confirm:

```
sudo apt-get install xrdp
```

Later, the installation starts.

A terminal window titled 'pi@raspberrypi: ~' with standard window controls. It shows the command 'sudo apt-get install xrdp' being executed. The output lists extra packages (vnc4server, x11-apps, etc.), suggested packages (vnc-java, mesa-utils, etc.), and NEW packages (vnc4server, x11-apps, etc., and xrdp). It also shows disk space requirements and a confirmation prompt '[Y/n] y' which has been answered with 'y'.

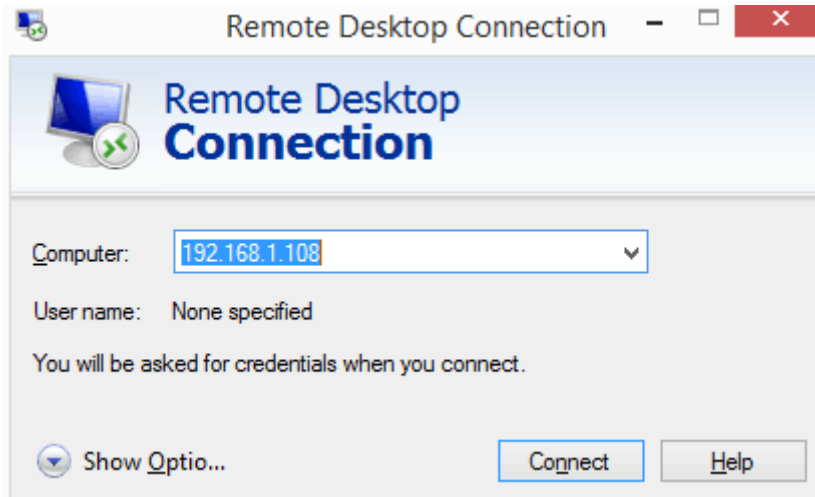
```
pi@raspberrypi:~ $ sudo apt-get install xrdp
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following extra packages will be installed:
  vnc4server x11-apps x11-session-utils xbase-clients xbitmaps xfonts-base
Suggested packages:
  vnc-java mesa-utils x11-xfs-utils
The following NEW packages will be installed:
  vnc4server x11-apps x11-session-utils xbase-clients xbitmaps xfonts-base
  xrdp
0 upgraded, 7 newly installed, 0 to remove and 0 not upgraded.
Need to get 8,468 kB of archives.
After this operation, 17.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Enter "Y", press key "Enter" to confirm.

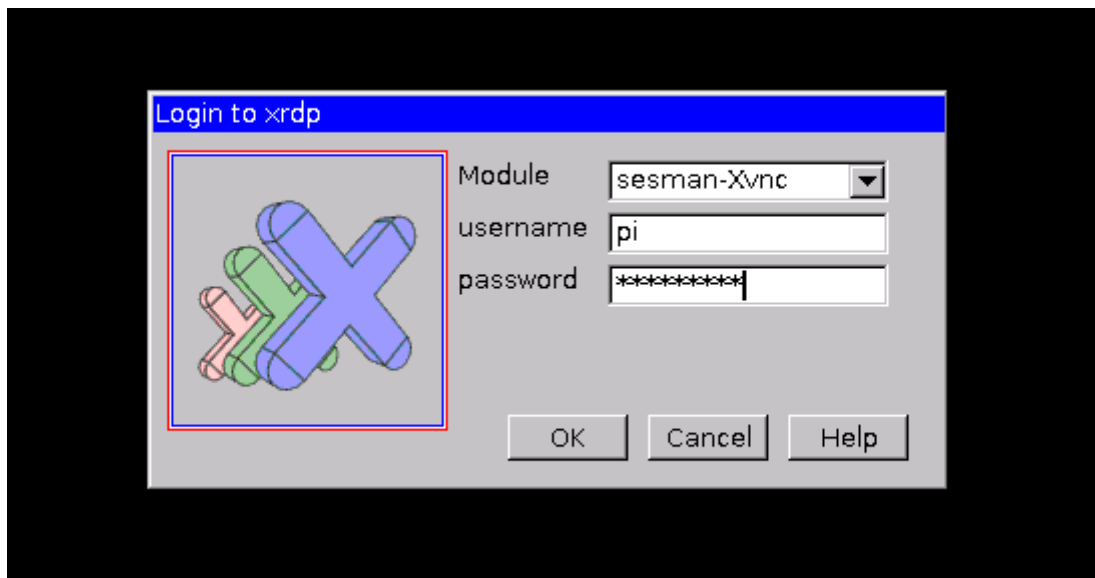
After the installation is completed, you can use Windows remote desktop applications to login to your RPi.

Login to Windows remote desktop

Use "WIN+R" or search function, open the remote desktop application "mstsc.exe" under Windows, enter the IP address of RPi and then click "Connect".



Later, there will be xrdp login screen. Enter the user name and password of RPi (RPi default user name: **pi**; password: **raspberrypi**) and click "OK".



Later, you can enter the RPi desktop system.



Here, you have successfully used the remote desktop login to RPi.

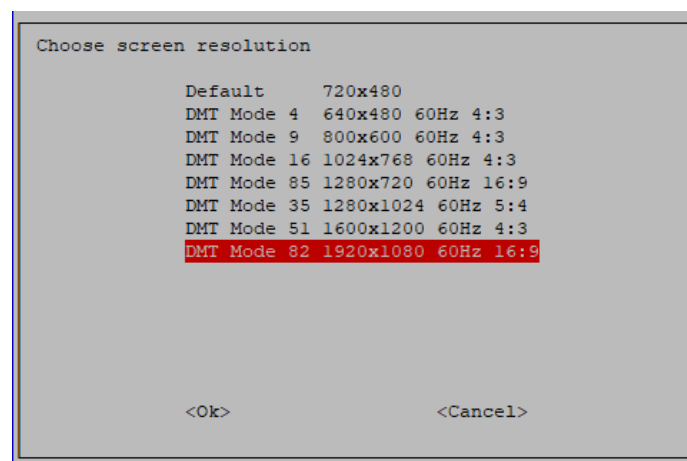
If you have any concerns, please feel free to contact us: support@freenove.com

You can set a proper resolution ratio according to your monitor.

And if VNC doesn't work normal, you need adjust resolution to try.

```
sudo raspi-config
```

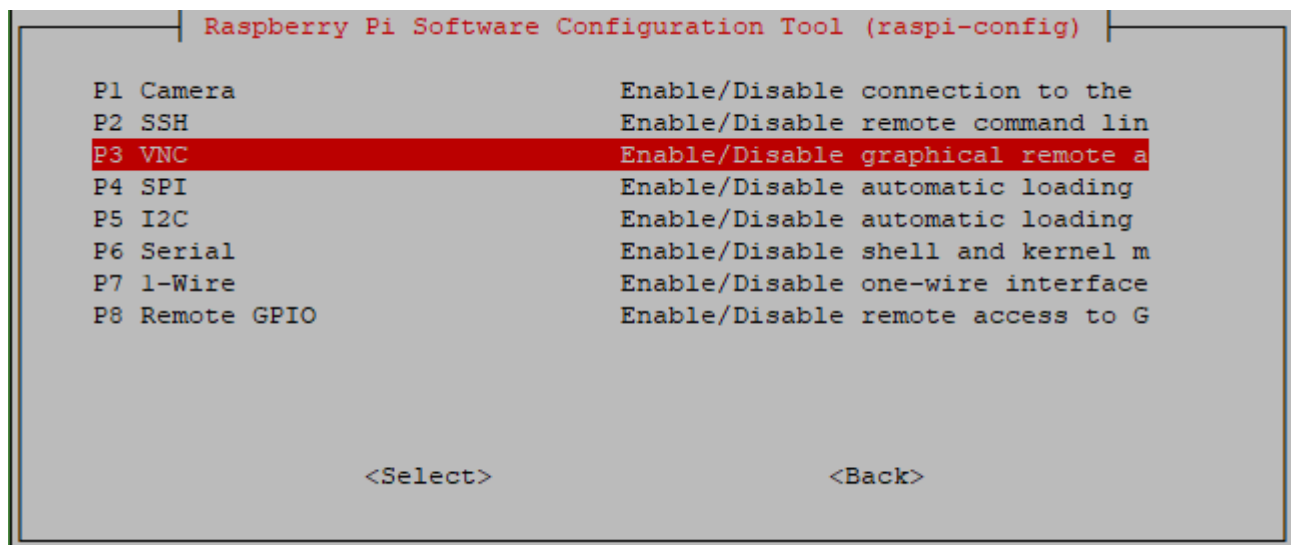
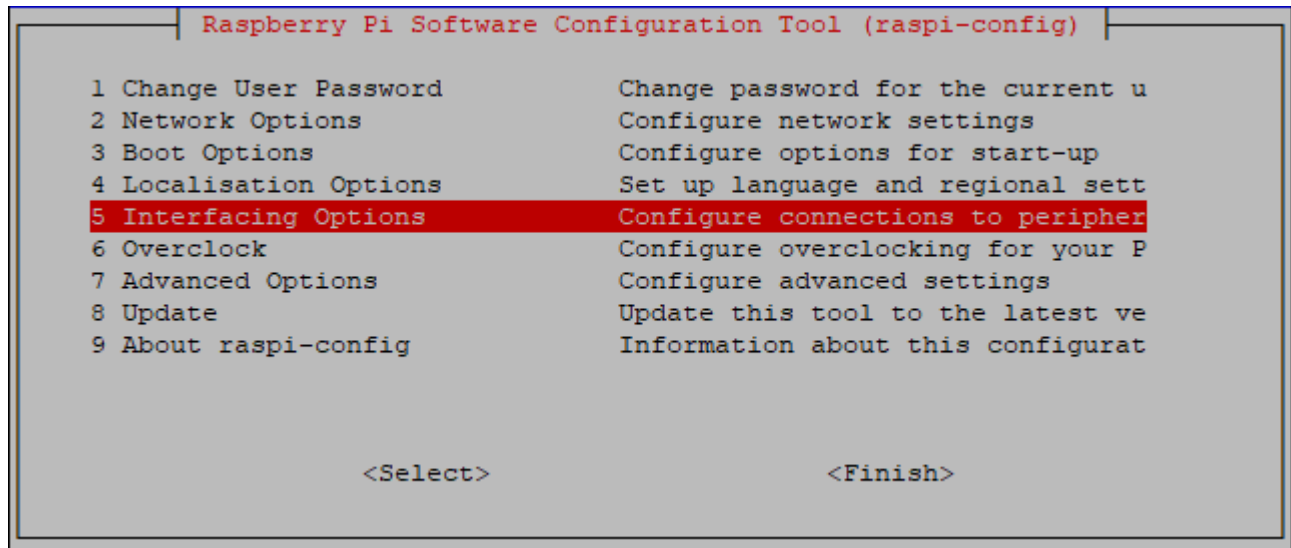
Select 7 Advanced Options→A5 Resolution→proper resolution ratio (set by yourself)→OK. If it needs restart, just restart.



VNC Viewer & VNC

Type the following command. And select 5 Interfacing Options→P3 VNC →Yes→OK→Finish. Here Raspberry Pi may need be restarted, and choose ok. Then open VNC interface.

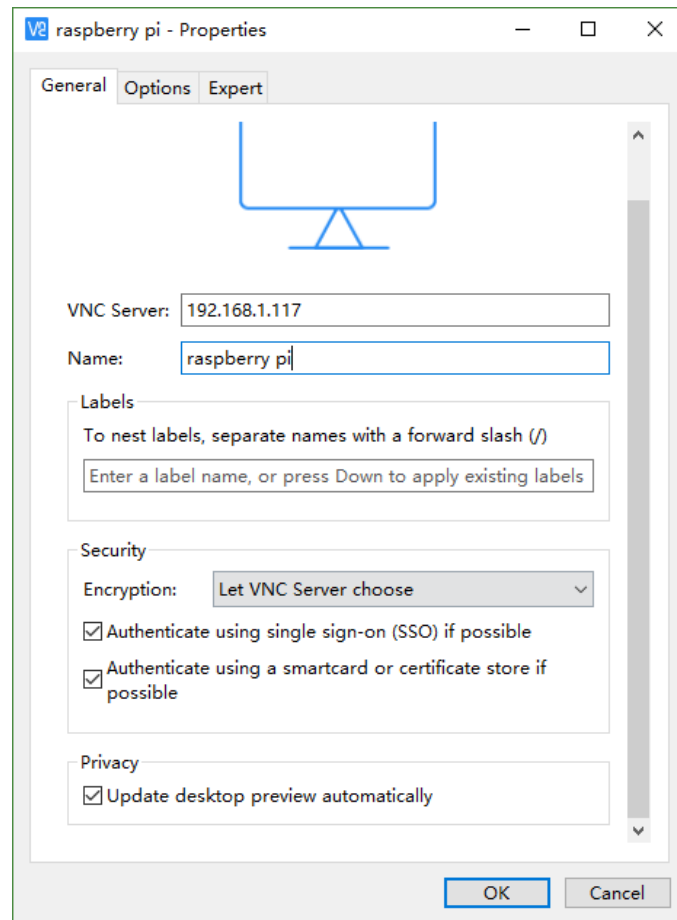
```
sudo raspi-config
```



Then download and install VNC Viewer by click following link:

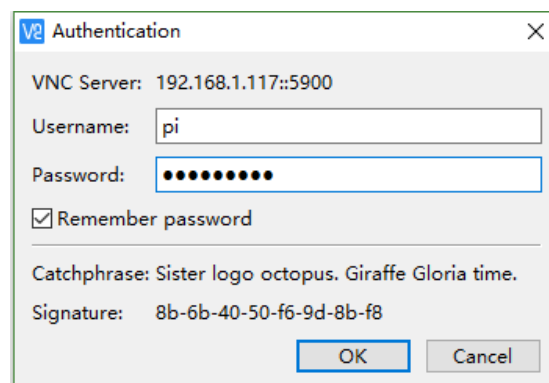
<https://www.realvnc.com/en/connect/download/viewer/windows/>

After installation is completed, open VNC Viewer. And click File → New Connection. Then the interface is shown below.



Enter IP address of your Raspberry Pi and fill in a Name. And click OK.

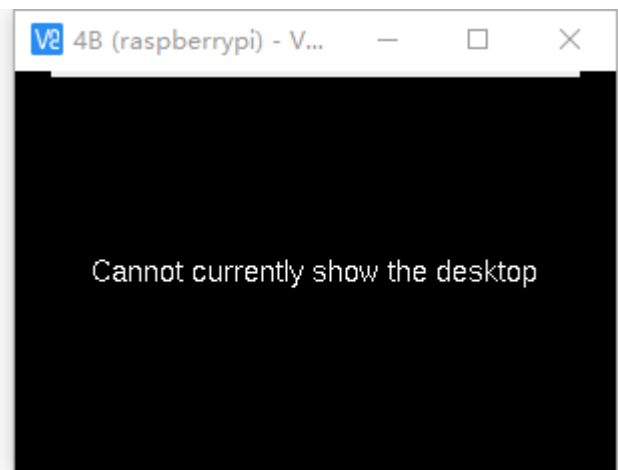
Then on the VNC Viewer panel, double-click new connection you just created, and the following dialog box pops up.



Enter username: pi and Password: raspberry. And click OK.



Here, you have logged in to Raspberry Pi successfully by using VNC Viewer
If the window does not work normal, as below.

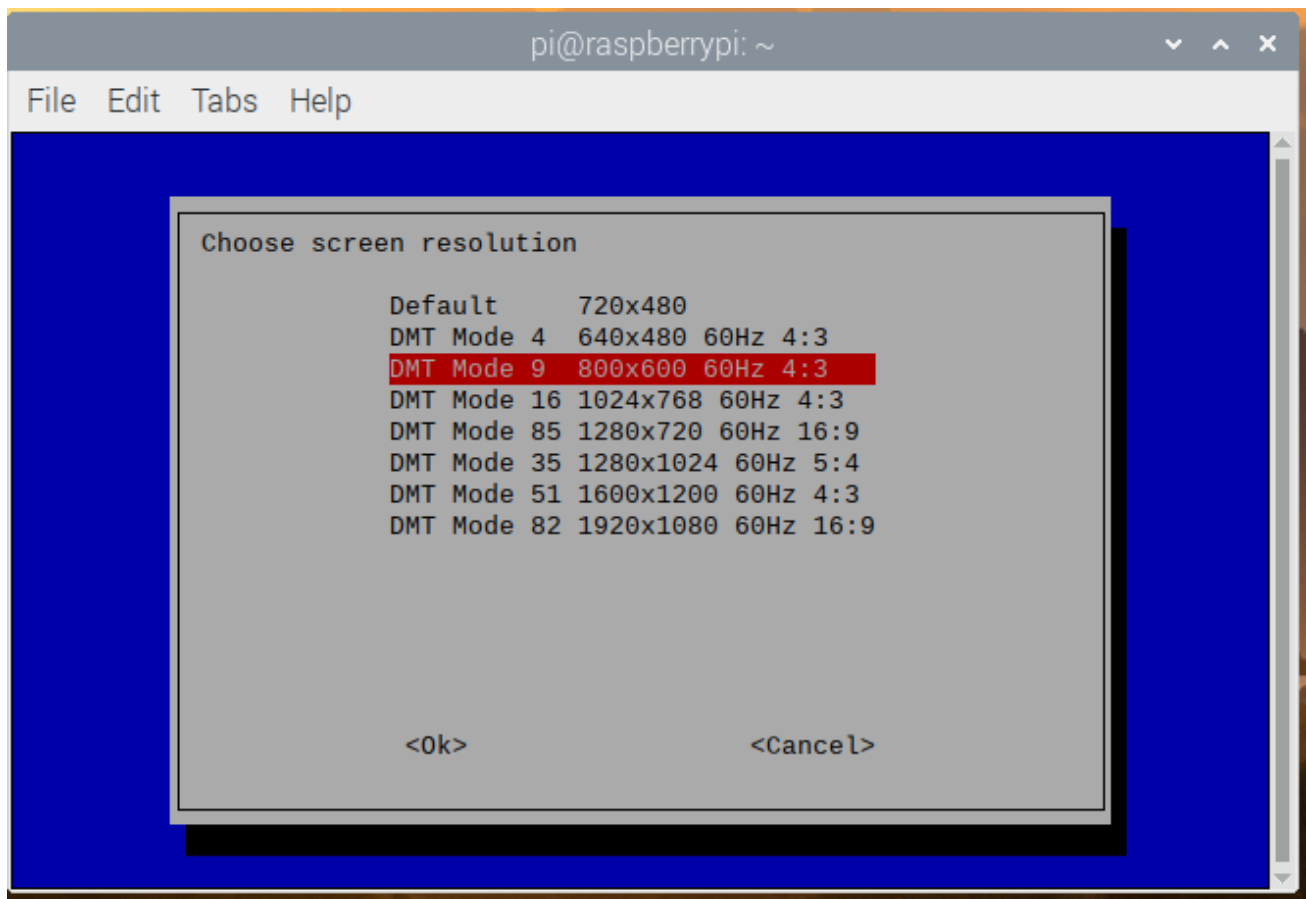


You need login to [windows remote desktop](#) and change resolution ratio.

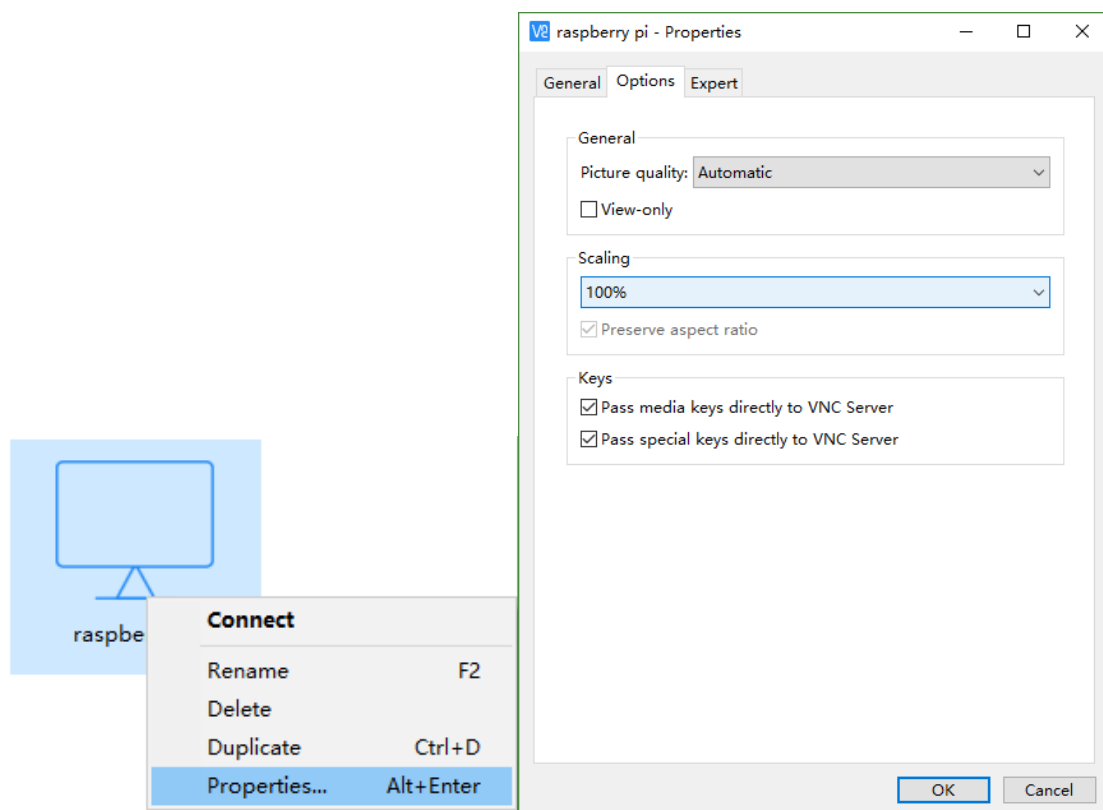
If you think resolution ratio is not OK, you can set a proper resolution ratio on set interface of Raspberry Pi.

```
sudo raspi-config
```

Select 7 Advanced Options → A5 Resolution → proper resolution ratio (set by yourself) → OK. If it needs restart, just restart.



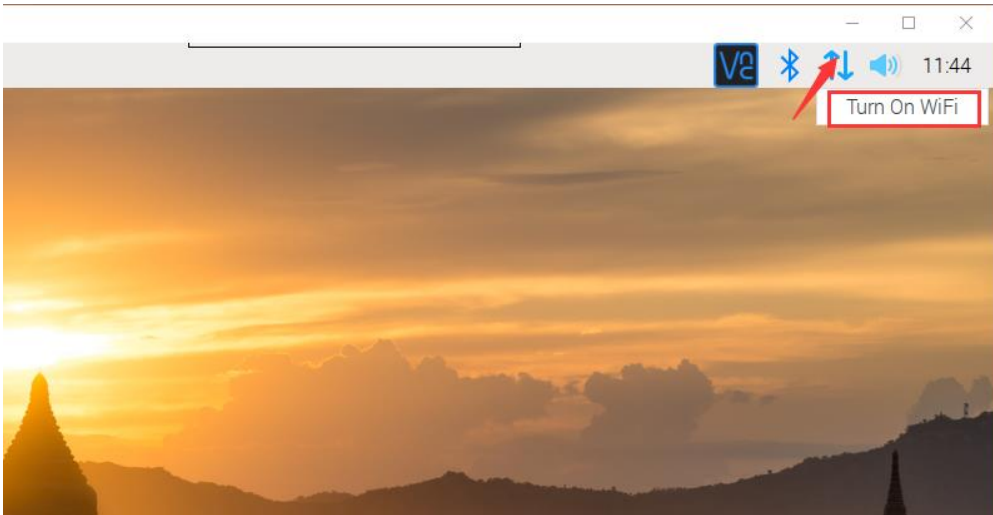
In addition, your VNC Viewer window may zoom your Raspberry Pi desktop. You can change it. On your VNC View control panel, click right key. And select Properties->Options label->Scaling. Then set proper scaling.



Here, you have logged in to Raspberry Pi successfully by using VNC Viewer and operated proper setting. Then continue to do some preparation work: install a GPIO library wiringPi for your RPi.

Wi-Fi

Raspberry Pi 4B/ 3B+/3B integrates a Wi-Fi adaptor. You can use it to connect to your Wi-Fi. Then you can use the wireless remote desktop to control your RPi. This will be helpful for the following work. Raspberry Pi of other models can use wireless remote desktop through accessing an external USB wireless card.



What's Next?

If you do not have Freenove project kit, you can visit official website to learn how to use Raspberry Pi.

<https://projects.raspberrypi.org/en/projects/raspberry-pi-using>

If you have **Freenove Ultimate Starter Kit for Raspberry Pi** or some electronic components, you can continue this tutorial to complete dozens of projects.

Part 2 Raspberry Pi Projects

This part is for Raspberry Pi projects.

1. If you already get **Freenove Ultimate Starter Kit for Raspberry Pi** or some electronic components, you can also download and learn the whole resource.

https://github.com/Freenove/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/archive/master.zip

It will teach you how to use raspberry Pi to control electronic components (such as LED, motor, servo, sensors) to create interesting projects. We provide code of both C and Python for each project, and code in Processing.pdf is based on Java.

2. We also have a **Raspberry Pi Car**. You can download the whole resource here.

https://github.com/Freenove/Freenove_Three-wheeled_Smart_Car_Kit_for_Raspberry_Pi/archive/master.zip

You can get Freenove Ultimate Starter Kit for Raspberry Pi, Freenove Three-wheeled Smart Car Kit for Raspberry Pi and other Freenove products here:

<http://www.freenove.com/store.html>

If you have any concerns, please feel free to contact us.

support@freenove.com

Following are tutorial examples of Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi.

Contents of Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi

Preface

- Raspberry Pi
- GPIO Extension Board
- Breadboard Power Module
- C code & Python code

Chapter 0 Preparation

- Step 0.1 Install the System
- Step 0.2 Install WiringPi
- Step 0.3 Obtain the Experiment Code
- Step 0.4 Code Editor
- Next

Chapter 1 LED

- Project 1.1 Blink

Chapter 2 Button & LED

- Project 2.1 Button & LED
- Project 2.2 MINI Table Lamp

Chapter 3 LED Bar Graph

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Chapter 4 Analog & PWM

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Chapter 5 RGB LED

- Project 5.1 Colorful LED

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- Project 6.1 Doorbell
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- Project 7.1 Read the Voltage of Potentiometer

Chapter 8 Potentiometer & LED

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Chapter 15 Servo

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Chapter 17 74HC595 & LEDBar Graph

Project 17.1 Flowing Water Light

Chapter 18 74HC595 & 7-segment display.

Project 18.1 7-segment display.

Project 18.2 4-Digit 7-segment display

Chapter 19 74HC595 & LED Matrix

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Chapter 20 LCD1602

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Project 24.1 Ultrasonic Ranging

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Project 26.1 Remote LED

Chapter 27 Solder Circuit Board

Project 27.1 Solder a Buzzer

Project 27.2 Solder a Flowing Water Light

What's next?

You can download the whole resource here:

https://github.com/Freenove/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/archive/master.zip

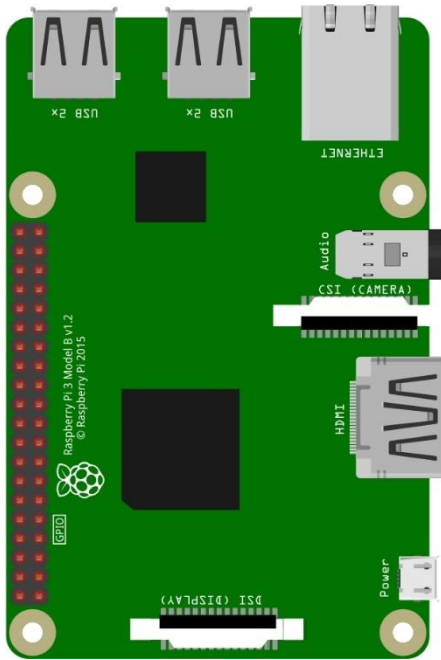
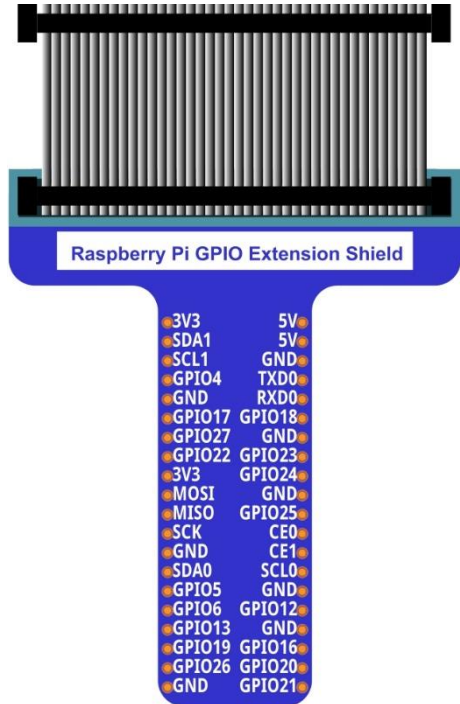
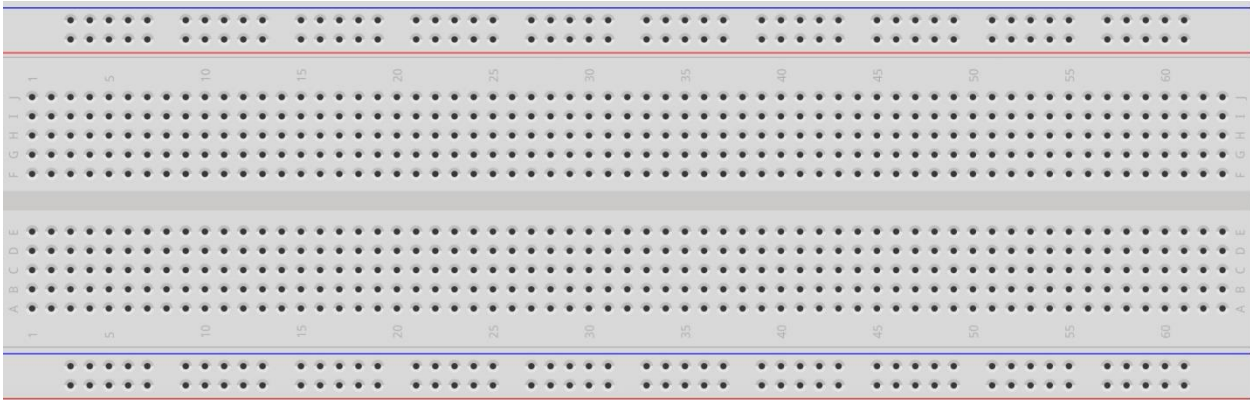
Chapter 1 LED (Sample)

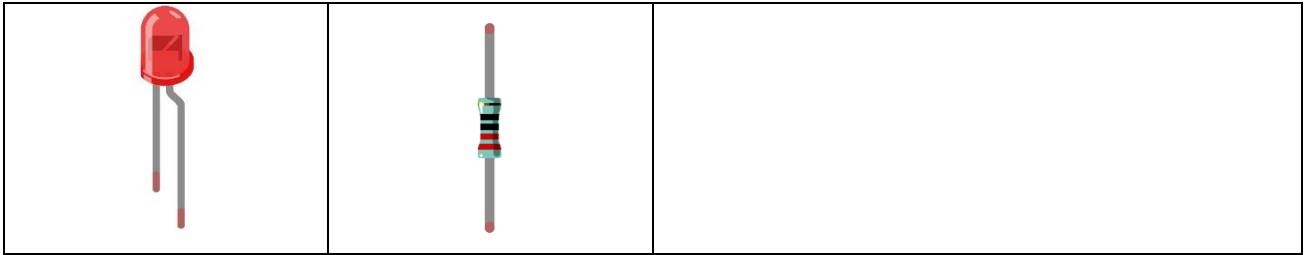
This chapter is the starting point of the journey to explore RPi electronic projects. Let's start with simple "Blink".

Project 1.1 Blink

In this project, let's try to use RPi to control LED blinking.

Component List

Raspberry Pi 3B x1 	GPIO Extension Board & Wire x1 	
BreadBoard x1 		
LED x1	Resistor 220Ω x1	Jumper 



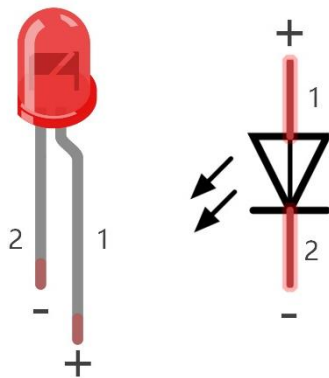
In the components list, 3B GPIO, Extension Shield Raspberry and Breadboard are necessary for each project. They will be listed only in text form later.

Component knowledge

LED

LED is a kind of diode. LED will shine only if the long pin of LED is connected to the positive electrode and the short pin is connected to negative electrode.

This is also the features of the common diode. Diode works only if the voltage of its positive electrode is higher than its negative electrode.



LED	Voltage	Maximum current	Recommended current
Red	1.9-2.2V	20mA	10mA
Green	2.9-3.4V	10mA	5mA
Blue	2.9-3.4V	10mA	5mA
Volt ampere characteristics conform to diode			

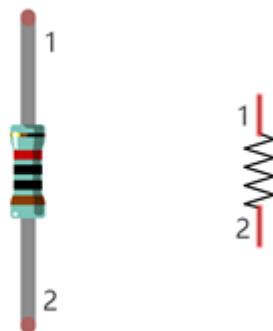
The LED can not be directly connected to power supply, which can damage component. A resistor with certain resistance must be connected in series in the circuit of LED.

Resistor

The unit of resistance(R) is ohm(Ω). $1\text{m}\Omega=1000\text{k}\Omega$, $1\text{k}\Omega=1000\Omega$.

Resistor is an electrical component that limits or regulates the flow of current in an electronic circuit.

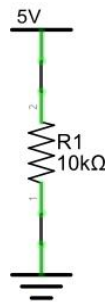
The left is the appearance of resistor, and the right is the symbol of resistor represented in circuit.



Color rings attached to the resistor is used to indicate its resistance. For more details of resistor color code, please refer to the appendix of this tutorial.

With the same voltage there will be less current with more resistance. And the links among current, voltage and resistance can be expressed by the formula below: $I=U/R$.

In the following diagram, the current through R1 is: $I=U/R=5V/10k\Omega=0.0005A=0.5mA$.



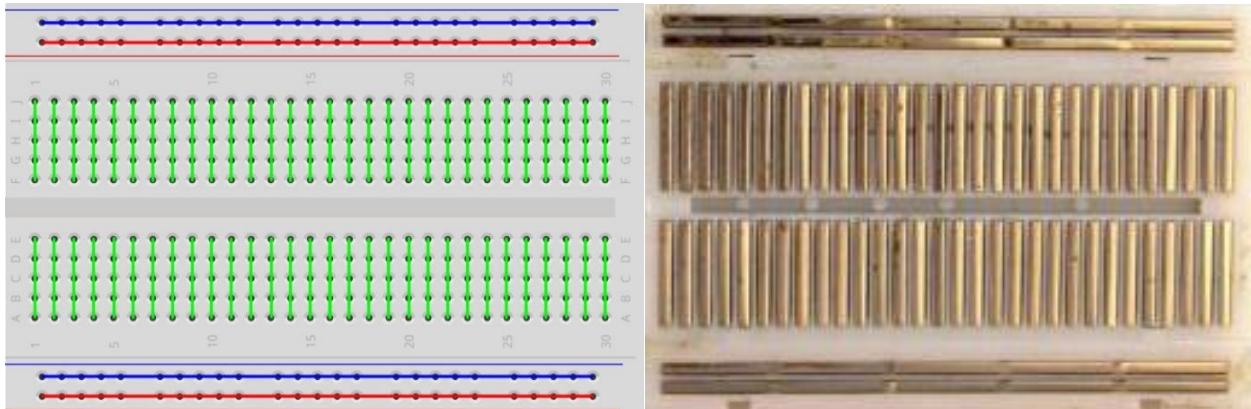
Do not connect the two poles of power supply with low resistance, which will make the current too high to damage electronic components.

And resistor has no poles.

Breadboard

Take a short breadboard as an example to introduce its feature, as below.

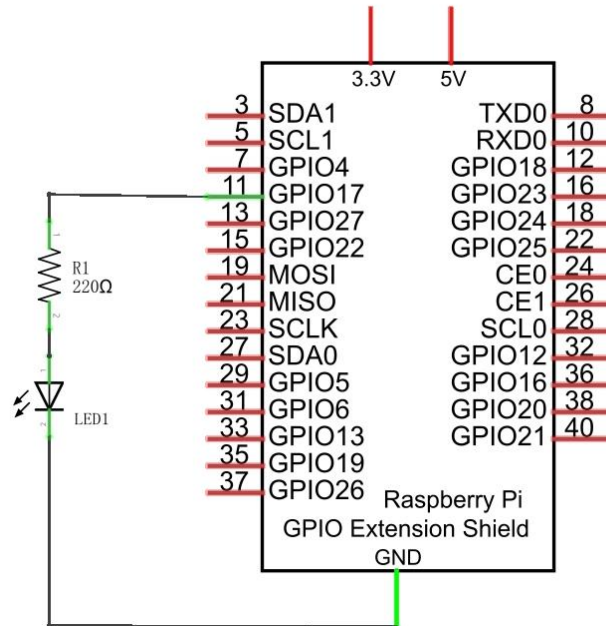
The left picture shows the way of connection of pins. The right picture shows the practical internal structure.



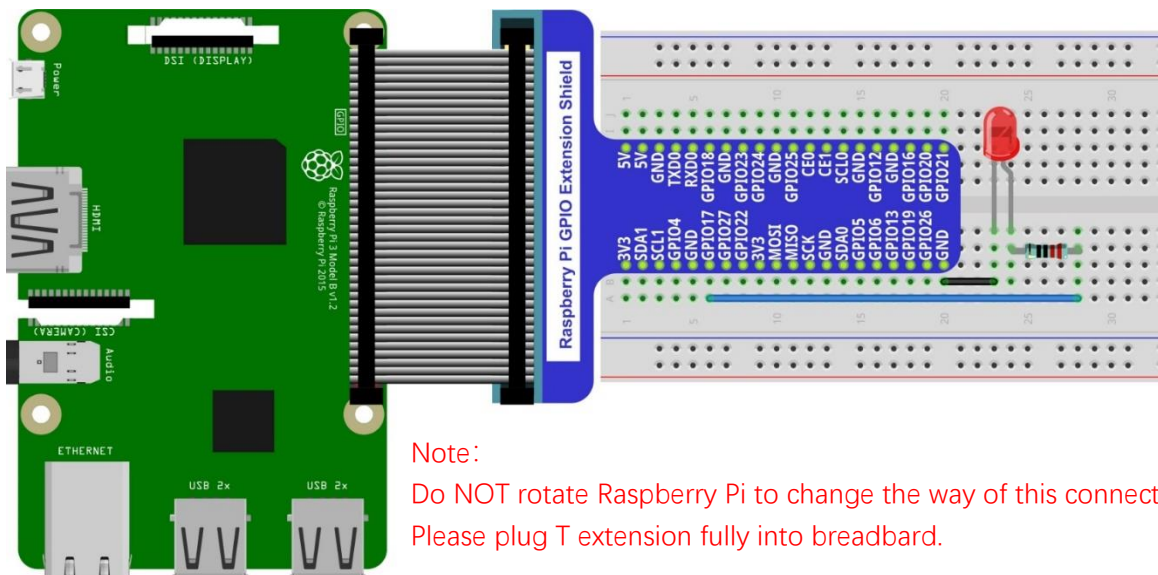
Circuit

Disconnect RPi from GPIO Extension Shield first. Then build the circuit according to the circuit diagram and the hardware connection diagram. After the circuit is built and confirmed, connect RPi to GPIO Extension Shield. In addition, short circuit (especially 5V and GND, 3.3V and GND) should be avoid, because short circuit may cause abnormal circuit work, or even damage to RPi.

Schematic diagram



Hardware connection. If you need any support, please feel free to contact us via: support@freenove.com



Because Numbering of GPIO Extension Shield is the same as RPi GPIO, later Hardware connection diagram will only show the part of breadboard and GPIO Extension Shield.

Code

According to the circuit, when the GPIO17 of RPi output high level, LED is turned on. Conversely, when the GPIO17 RPi output low level, LED is turned off. Therefore, we can let GPIO17 output high and low level in

cycle to make LED blink. We will use both C code and Python code to achieve the target.

C Code 1.1.1 Blink

First, execute command into the terminal one by one. Then observe the project result, and analyze the code.

1. Use cd command to enter 01.1.1_Blink directory of C code.

```
cd ~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/C_Code/01.1.1_Blink
```

2. Use the following command to compile the code "Blink.c" and generate executable file "Blink".

"l" of "lwiringPi" is low case of "L".

```
gcc Blink.c -o Blink -lwiringPi
```

3. Then run the generated file "blink".

```
sudo ./Blink
```

Now, LED start blink.

```
pi@raspberrypi: ~
File Edit Tabs Help
pi@raspberrypi:~ $ cd ~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/C_Code/01.1.1_Blink
pi@raspberrypi:~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/C_Code/01.1.1_Blink $ gcc Blink.c -o Blink -lwiringPi
pi@raspberrypi:~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/C_Code/01.1.1_Blink $ sudo ./Blink
wiringPi initialize successfully, GPIO 0(wiringPi pin)
led on...
...led off
led on...
```

You can press "Ctrl+C" to end the program. The following is the program code:

```
1  #include <wiringPi.h>
2  #include <stdio.h>
3
4  #define ledPin    0
5
6  int main(void)
7  {
8      if(wiringPiSetup() == -1){ //when initialize wiring failed, print message to screen
9          printf("setup wiringPi failed !");
10         return 1;
11     }
12     //when initialize wiring successfully, print message to screen
13     printf("wiringPi initialize successfully, GPIO %d(wiringPi pin)\n",ledPin);
14
15     pinMode(ledPin, OUTPUT);
16
17     while(1){
18         digitalWrite(ledPin, HIGH); //led on
19         printf("led on...\n");
20         delay(1000);
21         digitalWrite(ledPin, LOW); //led off
```

```

22         printf("...led off\n");
23         delay(1000);
24     }
25
26     return 0;
27 }

```

GPIO connected to ledPin in the circuit is GPIO17. And GPIO17 is defined as 0 in the wiringPi numbering. So ledPin should be defined as 0 pin. You can refer to the corresponding table in Chapter 0.

```
#define ledPin 0
```

In the main function main(), initialize wiringPi first, and then print out the initial results. Once the initialization fails, exit the program.

```

if(wiringPiSetup() == -1){ //when initialize wiring failed, print message to screen
    printf("setup wiringPi failed !");
    return 1;
}
//when initialize wiring successfully, print message to screen
printf("wiringPi initialize successfully, GPIO %d(wiringPi pin)\n", ledPin);

```

After the wiringPi is initialized successfully, set the ledPin to output mode. And then enter the while cycle, which is an endless loop. That is, the program will always be executed in this cycle, unless it is ended outside. In this cycle, use digitalWrite (ledPin, HIGH) to make ledPin output high level, then LED is turned on. After a period of time delay, use digitalWrite(ledPin, LOW) to make ledPin output low level, then LED is turned off, which is followed by a delay. Repeat the cycle, then LED will start blinking.

```

pinMode(ledPin, OUTPUT);
while(1){
    digitalWrite(ledPin, HIGH); //led is turned on
    printf("led on...\n");
    delay(1000);
    digitalWrite(ledPin, LOW); //led is turned off
    printf("...led off\n");
    delay(1000);
}

```

Among them, the configuration function for GPIO is shown below as:

```
void pinMode(int pin, int mode);
```

This sets the mode of a pin to either INPUT, OUTPUT, PWM_OUTPUT or GPIO_CLOCK. Note that only wiringPi pin 1 (BCM_GPIO 18) supports PWM output and only wiringPi pin 7 (BCM_GPIO 4) supports CLOCK output modes.

This function has no effect when in Sys mode. If you need to change the pin mode, then you can do it with the gpio program in a script before you start your program

```
void digitalWrite (int pin, int value);
```

Writes the value HIGH or LOW (1 or 0) to the given pin which must have been previously set as an output.

For more related functions, please refer to <http://wiringpi.com/reference/>

Python Code 1.1.1 Blink

Net, we will use Python language to make LED blink.

First, observe the project result, and then analyze the code.

1. Use cd command to enter 01.1.1_Blink directory of Python code.

```
cd ~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/Python_Code/01.1.1_Blink
```

2. Use python command to execute python code blink.py.

```
python Blink.py
```

Now, LED start blinking.

```
pi@raspberrypi:~ $ cd ~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/Python_Code/01.1.1_Blink
pi@raspberrypi:~/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/Code/Python_Code/01.1.1_Blink $ python Blink.py
```

You can press "Ctrl+C" to end the program. The following is the program code:

```

1  import RPi.GPIO as GPIO
2  import time
3
4  ledPin = 11    # RPI Board pin11
5
6  def setup():
7      GPIO.setmode(GPIO.BOARD)      # Numbers GPIOs by physical location
8      GPIO.setup(ledPin, GPIO.OUT)   # Set ledPin's mode is output
9      GPIO.output(ledPin, GPIO.LOW)  # Set ledPin low to off led
10     print ('using pin%d'%ledPin)
11
12     def loop():
13         while True:
14             GPIO.output(ledPin, GPIO.HIGH) # led on
15             print ('...led on')
16             time.sleep(1)                  # delay 1 second
17             GPIO.output(ledPin, GPIO.LOW)  # led off
18             print ('led off...')
19             time.sleep(1)
20     def destroy():
21         GPIO.output(ledPin, GPIO.LOW)      # led off
22         GPIO.cleanup()                     # Release resource
23
24     if __name__ == '__main__':             # Program start from here
25         setup()
26         try:
27             loop()
28         except KeyboardInterrupt: # When 'Ctrl+C' is pressed, the subprogram destroy() will
29             be executed.
30             destroy()
31
```

In subfunction setup(), GPIO.setmode (GPIO.BOARD) is used to set the serial number for GPIO based on

physical location of the pin. GPIO17 use pin 11 of the board, so define ledPin as 11 and set ledPin to output mode (output low level).

```
ledPin = 11    # RPi Board pin11
def setup():
    GPIO.setmode(GPIO.BOARD)      # Numbers GPIOs by physical location
    GPIO.setup(ledPin, GPIO.OUT)   # Set ledPin to output mode
    GPIO.output(ledPin, GPIO.LOW) # Set ledPin to low level to turn off led
    print ('using pin%d'%ledPin)
```

In loop(), there is a while cycle, which is an endless loop. That is, the program will always be executed in this cycle, unless it is ended outside. In this cycle, set ledPin output high level, then LED is turned on. After a period of time delay, set ledPin output low level, then LED is turned off, which is followed by a delay. Repeat the cycle, then LED will start blinking.

```
def loop():
    while True:
        GPIO.output(ledPin, GPIO.HIGH) # led on
        print ('... led on')
        time.sleep(1)
        GPIO.output(ledPin, GPIO.LOW) # led off
        print ('led off...')
        time.sleep(1)
```

Finally, when the program is terminated, subfunction will be executed, the LED will be turned off and then the IO port will be released. If close the program terminal directly, the program will be terminated too, but destroy() function will not be executed. So, GPIO resources won't be released, in the warning message may appear next time you use GPIO. So, it is not a good habit to close the program terminal directly.

```
def destroy():
    GPIO.output(ledPin, GPIO.LOW)    # led is turned off
    GPIO.cleanup()                  # Release resource
```

About RPi.GPIO:

RPi.GPIO

This is a Python module to control the GPIO on a Raspberry Pi. It includes basic output function and input function of GPIO, and function used to generate PWM.

GPIO.setmode(mode)

Set the mode for pin serial number of GPIO.

mode=GPIO.BOARD, which represents the GPIO pin serial number is based on physical location of RPi.

mode=GPIO.BCM, which represents the pin serial number is based on CPU of BCM chip.

GPIO.setup(pin, mode)

Set pin to input mode or output mode. "pin" for the GPIO pin, "mode" for INPUT or OUTPUT.

GPIO.output(pin, mode)

Set pin to output mode. "pin" for the GPIO pin, "mode" for HIGH (high level) or LOW (low level).

For more functions related to RPi.GPIO, please refer to:

<https://sourceforge.net/p/raspberry-gpio-python/wiki/Examples/>

"import time" time is a module of python.

<https://docs.python.org/2/library/time.html?highlight=time%20time#module-time>

You can download the whole resource here:

https://github.com/Freenove/Freenove_Ultimate_Starter_Kit_for_Raspberry_Pi/archive/master.zip

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